

Important Questions for Class 12

Maths

Chapter 2 – Inverse Trigonometric Functions

Very Short Answer Questions.

1 Mark

1. Write the principal value

(i). Write the principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$.

Ans: Let $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = x$

Taking sine on both sides we get,

$$\sin(x) = -\frac{\sqrt{3}}{2}$$

But, the principal value of $\sin\left(\frac{\pi}{3}\right)$ is $\frac{\sqrt{3}}{2}$ and we know that and

$$\sin(-x) = -\sin(x).$$

$$\Rightarrow \sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$\therefore x = -\frac{\pi}{3}$$

Hence, the principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ is $-\frac{\pi}{3}$.

(ii). Write the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

Ans: Let $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = x$

Taking cosine on both sides we get,

$$\cos(x) = \frac{\sqrt{3}}{2}$$

But, the principal value of $\cos\left(\frac{\pi}{6}\right)$ is $\frac{\sqrt{3}}{2}$.

$$\therefore x = \frac{\pi}{6}$$

Hence, the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is $\frac{\pi}{6}$.

(iii). Write the principal value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$.

Ans: Let $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = x$

Taking tan on both sides we get,

$$\tan(x) = -\frac{1}{\sqrt{3}}$$

But, the principal value of $\tan\left(\frac{\pi}{6}\right)$ is $\frac{1}{\sqrt{3}}$ and we know that and

$$\tan(-x) = -\tan(x).$$

$$\Rightarrow \tan\left(-\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}}$$

$$\therefore x = -\frac{\pi}{6}$$

Hence, the principal value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ is $-\frac{\pi}{6}$.

(iv). Write the principal value of $\operatorname{cosec}^{-1}(-2)$.

Ans: Let $\operatorname{cosec}^{-1}(-2) = x$

Taking cosec on both sides we get,

$$\operatorname{cosec}(x) = -2$$

But, the principal value of $\sin\left(\frac{\pi}{6}\right)$ is $\frac{1}{2}$ i.e., the principal value of $\operatorname{cosec}\left(\frac{\pi}{6}\right)$ is

2 and we know that and $\operatorname{cosec}(-x) = -\operatorname{cosec}(x)$.

$$\Rightarrow \operatorname{cosec}\left(-\frac{\pi}{6}\right) = -2$$

$$\therefore x = -\frac{\pi}{6}$$

Hence, the principal value of $\operatorname{cosec}^{-1}(-2)$ is $-\frac{\pi}{6}$.

(v). Write the principal value of $\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)$.

Ans: Let $\cot^{-1}\left(\frac{1}{\sqrt{3}}\right) = x$

Taking cot on both sides we get,

$$\cot(x) = \frac{1}{\sqrt{3}}$$

But, the principal value of $\tan\left(\frac{\pi}{3}\right)$ is $\sqrt{3}$ i.e., the principal value of $\cot\left(\frac{\pi}{3}\right)$ is

$$\frac{1}{\sqrt{3}}.$$

$$\Rightarrow \cot\left(\frac{\pi}{3}\right) = \frac{1}{\sqrt{3}}$$

$$\therefore x = \frac{\pi}{3}$$

Hence, the principal value of $\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)$ is $\frac{\pi}{3}$.

(vi). Write the principal value of $\sec^{-1}(-2)$.

Ans: Let $\sec^{-1}(-2) = x$

Taking sec on both sides we get, $\sec(x) = -2$

But, the principal value of $\cos\left(\frac{2\pi}{3}\right)$ is $-\frac{1}{2}$ i.e., the principal value of $\sec\left(\frac{2\pi}{3}\right)$ is -2 .

$$\Rightarrow \sec\left(\frac{2\pi}{3}\right) = -2$$

$$\therefore x = \frac{2\pi}{3}$$

Hence, the principal value of $\sec^{-1}(-2)$ is $\frac{2\pi}{3}$.

(vii). Write the principal value of

$$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \cos^{-1}\left(-\frac{1}{2}\right) + \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right).$$

Ans: Let $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = x$

Taking sine on both sides we get,

$$\sin(x) = -\frac{\sqrt{3}}{2}$$

But, the principal value of $\sin\left(\frac{\pi}{3}\right)$ is $\frac{\sqrt{3}}{2}$ and we know that and

$$\sin(-x) = -\sin(x)$$

$$\Rightarrow \sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$\therefore x = -\frac{\pi}{3}$$

Let $\cos^{-1}\left(-\frac{1}{2}\right) = y$

Taking cosine on both sides we get,

$$\cos(y) = -\frac{1}{2}$$

But, the principal value of $\cos\left(\frac{2\pi}{3}\right)$ is $-\frac{1}{2}$.

$$\Rightarrow \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$$

$$\therefore y = \frac{2\pi}{3}$$

Let $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = z$

Taking tan on both sides we get,

$$\tan(z) = -\frac{1}{\sqrt{3}}$$

But, the principal value of $\tan\left(\frac{\pi}{6}\right)$ is $\frac{1}{\sqrt{3}}$ and we know that and

$$\tan(-x) = -\tan(x).$$

$$\Rightarrow \tan\left(-\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}}$$

$$\therefore z = -\frac{\pi}{6}$$

Hence, the principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \cos^{-1}\left(-\frac{1}{2}\right) + \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ is

$x + y + z$.

$$\Rightarrow -\frac{\pi}{3} + \frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}.$$

2. Write the value of the function.

(i). What is the value of the function $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) - \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$.

Ans: Let $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = x$

Taking tan on both sides we get,

$$\tan(x) = \frac{1}{\sqrt{3}}$$

But, the principal value of $\tan\left(\frac{\pi}{6}\right)$ is $\frac{1}{\sqrt{3}}$.

$$\Rightarrow \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$$

$$\therefore x = \frac{\pi}{6}$$

Let $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = y$

Taking sec on both sides we get,

$$\sec(y) = \frac{2}{\sqrt{3}}$$

But, the principal value of $\cos\left(\frac{\pi}{6}\right)$ is $\frac{\sqrt{3}}{2}$

$$\Rightarrow \sec\left(\frac{\pi}{6}\right) = \frac{2}{\sqrt{3}}$$

$$\therefore y = \frac{\pi}{6}$$

Hence, the principal value of $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) - \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ is $x + y$.

$$\Rightarrow \frac{\pi}{6} - \frac{\pi}{6} = 0.$$

(ii). What is the value of the function $\sin^{-1}\left(-\frac{1}{2}\right) - \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

Ans: Let $\sin^{-1}\left(-\frac{1}{2}\right) = x$

Taking sine on both sides we get,

$$\sin(x) = -\frac{1}{2}$$

But, the principal value of $\sin\left(\frac{\pi}{6}\right)$ is $\frac{1}{2}$ and we know that $\sin(-x) = -\sin(x)$.

$$\Rightarrow \sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$$

$$\therefore x = -\frac{\pi}{6}$$

Let $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = y$

Taking cos on both sides we get,

$$\cos(y) = \frac{\sqrt{3}}{2}$$

But, the principal value of $\cos\left(\frac{\pi}{6}\right)$ is $\frac{\sqrt{3}}{2}$

$$\Rightarrow \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

$$\therefore y = \frac{\pi}{6}$$

Hence, the principal value of $\sin^{-1}\left(-\frac{1}{2}\right) - \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ is $x - y$.

$$\Rightarrow -\frac{\pi}{6} - \frac{\pi}{6} = -\frac{\pi}{3}.$$

(iii). What is the value of the function $\tan^{-1}(1) - \cot^{-1}(-1)$.

Ans: Let $\tan^{-1}(1) = x$

Taking tan on both sides we get, $\tan(x) = 1$

But, the principal value of $\tan\left(\frac{\pi}{4}\right)$ is 1.

$$\Rightarrow \tan\left(\frac{\pi}{4}\right) = 1$$

$$\therefore x = \frac{\pi}{4}$$

Let $\cot^{-1}(-1) = y$

Taking cot on both sides we get, $\cot(y) = -1$

But we know that principal value of $\cot\left(\frac{3\pi}{4}\right) = -1$.

$$\Rightarrow \cot\left(\frac{3\pi}{4}\right) = -1$$

$$\therefore y = \frac{3\pi}{4}$$

Hence, the principal value of $\tan^{-1}(1) - \cot^{-1}(-1)$ is $x - y$

$$\Rightarrow \frac{\pi}{4} - \frac{3\pi}{4} = -\frac{\pi}{2}$$

(iv). What is the value of the function $\operatorname{cosec}^{-1}(\sqrt{2}) + \sec^{-1}(\sqrt{2})$.

Ans: We know that the principal value of $\sin\left(\frac{\pi}{4}\right), \cos\left(\frac{\pi}{4}\right)$ is $\frac{1}{\sqrt{2}}$.

Let $\operatorname{cosec}^{-1}(\sqrt{2}) = x$ and $\sec^{-1}(\sqrt{2}) = y$

Taking cosec on both sides of first equation and sec on both sides of second equation we get, $\operatorname{cosec}(x) = \sqrt{2}$ and $\sec(y) = \sqrt{2}$.

But, the principal value of $\sin\left(\frac{\pi}{4}\right), \cos\left(\frac{\pi}{4}\right)$ is $\frac{1}{\sqrt{2}}$ and

$$\operatorname{cosec}(x) = \frac{1}{\sin x}, \sec(x) = \frac{1}{\cos x}$$

$$\operatorname{cosec}\left(\frac{\pi}{4}\right) = \sqrt{2} \text{ and } \sec\left(\frac{\pi}{4}\right) = \sqrt{2}$$

$$\therefore x = y = \frac{\pi}{4}$$

Hence, the principal value of $\operatorname{cosec}^{-1}(\sqrt{2}) + \sec^{-1}(\sqrt{2})$ is $x + y$.

$$\Rightarrow \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$$

(v). What is the value of the function $\tan^{-1}(1) + \cot^{-1}(1) + \sin^{-1}(1)$.

Ans: Let $\tan^{-1}(1) = x$

Taking tan on both sides we get, $\tan(x) = 1$

But, the principal value of $\tan\left(\frac{\pi}{4}\right)$ is 1.

$$\Rightarrow \tan\left(\frac{\pi}{4}\right) = 1$$

$$\therefore x = \frac{\pi}{4}$$

Let $\cot^{-1}(1) = y$

Using the trigonometric identity $\cot^{-1}(x) = \frac{\pi}{2} - \tan^{-1}(x)$ we get

$$y = \frac{\pi}{2} - \tan^{-1}(1)$$

But, $\tan^{-1}(1) = \frac{\pi}{4}$ (proved above)

$$\therefore y = \frac{\pi}{4}$$

Let $\sin^{-1}(1) = z$

Taking sine on both sides we get, $\sin(z) = 1$

But, the principal value of $\sin\left(\frac{\pi}{2}\right)$ is 1.

$$\Rightarrow \sin\left(\frac{\pi}{2}\right) = 1$$

$$\therefore z = \frac{\pi}{2}$$

Hence, the principal value of $\tan^{-1}(1) + \cot^{-1}(1) + \sin^{-1}(1)$ is $x + y + z$.

$$\Rightarrow \frac{\pi}{4} + \frac{\pi}{4} + \frac{\pi}{2} = \pi.$$

(vi). What is the value of the function $\sin^{-1}\left(\sin \frac{4\pi}{5}\right)$.

Ans: Let $\sin^{-1}\left(\sin \frac{4\pi}{5}\right) = x$

Taking sine on both sides we get, $\sin \frac{4\pi}{5} = \sin x$.

Writing $\sin\left(\frac{4\pi}{5}\right)$ as $\sin\left(\pi - \frac{\pi}{5}\right) = \sin\left(\frac{\pi}{5}\right)$.

$$\therefore x = \frac{\pi}{5}$$

Hence, the principal value of $\sin^{-1}\left(\sin \frac{4\pi}{5}\right)$ is $\frac{\pi}{5}$.

(vii). What is the value of the function $\tan^{-1}\left(\tan \frac{5\pi}{6}\right)$.

Ans: Let $\tan^{-1}\left(\tan \frac{5\pi}{6}\right) = x$

Taking tan on both sides we get, $\tan \frac{5\pi}{6} = \tan x$

Writing $\tan\left(\frac{5\pi}{6}\right)$ as $\tan\left(\pi - \frac{\pi}{6}\right) = -\tan\left(\frac{\pi}{6}\right)$. Also, we know that

$$\tan(-x) = -\tan(x).$$

$$\therefore x = -\frac{\pi}{6}$$

Hence, the principal value of $\tan^{-1}\left(\tan \frac{5\pi}{6}\right)$ is $-\frac{\pi}{6}$.

(viii). What is the value of the function $\operatorname{cosec}^{-1}\left(\operatorname{cosec} \frac{3\pi}{4}\right)$.

Ans: Let $\operatorname{cosec}^{-1}\left(\operatorname{cosec} \frac{3\pi}{4}\right)$

Taking cosec on both sides we get, $\operatorname{cosec} \frac{3\pi}{4} = \operatorname{cosec}(x)$

Writing $\operatorname{cosec}\left(\frac{3\pi}{4}\right)$ as $\operatorname{cosec}\left(\pi - \frac{\pi}{4}\right) = \operatorname{cosec}\left(\frac{\pi}{4}\right)$.

$$\therefore x = \frac{\pi}{4}$$

Hence, the principal value of $\operatorname{cosec}^{-1}\left(\operatorname{cosec}\frac{3\pi}{4}\right)$ is $\frac{\pi}{4}$.

Short Answer Questions

4 Marks

3. Show that $\tan^{-1}\left(\frac{\sqrt{1+\cos x} + \sqrt{1-\cos x}}{\sqrt{1+\cos x} - \sqrt{1-\cos x}}\right) = \frac{\pi}{4} + \frac{x}{2}$, $x \in [0, \pi]$.

Ans: Using the trigonometric identities

$\sqrt{1+\cos x} = 2\cos\left(\frac{x}{2}\right)$ and $\sqrt{1-\cos x} = 2\sin\left(\frac{x}{2}\right)$ we get,

$$\tan^{-1}\left(\frac{\sqrt{1+\cos x} + \sqrt{1-\cos x}}{\sqrt{1+\cos x} - \sqrt{1-\cos x}}\right) = \tan^{-1}\left(\frac{2\cos\left(\frac{x}{2}\right) + 2\sin\left(\frac{x}{2}\right)}{2\cos\left(\frac{x}{2}\right) - 2\sin\left(\frac{x}{2}\right)}\right) \dots\dots(1)$$

Dividing the numerator and denominator of (1) by $2\cos\left(\frac{x}{2}\right)$ we get,

$$\tan^{-1}\left(\frac{\sqrt{1+\cos x} + \sqrt{1-\cos x}}{\sqrt{1+\cos x} - \sqrt{1-\cos x}}\right) = \tan^{-1}\left(\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right) \dots\dots(2)$$

Now using the identity, $\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(x) + \tan^{-1}(y)$ on equation (2) we get,

$$\begin{aligned} \tan^{-1}\left(\frac{\sqrt{1+\cos x} + \sqrt{1-\cos x}}{\sqrt{1+\cos x} - \sqrt{1-\cos x}}\right) &= \tan^{-1}(1) + \tan^{-1}\left(\tan\left(\frac{x}{2}\right)\right) \\ \Rightarrow \frac{\pi}{4} + \frac{x}{2} \end{aligned}$$

4. Show that $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right) - \cot^{-1}\left(\sqrt{\frac{1+\cos x}{1-\cos x}}\right) = \frac{\pi}{4}$, $x \in \left(0, \frac{\pi}{2}\right)$.

Ans: Using the trigonometric identities

$\sqrt{1 + \cos x} = 2 \cos\left(\frac{x}{2}\right)$ and $\sqrt{1 - \cos x} = 2 \sin\left(\frac{x}{2}\right)$ on $\cot^{-1}\left(\sqrt{\frac{1 + \cos x}{1 - \cos x}}\right)$ we get,

$$\cot^{-1}\left(\sqrt{\frac{1 + \cos x}{1 - \cos x}}\right) = \cot^{-1}\left(\frac{2 \cos\left(\frac{x}{2}\right)}{2 \sin\left(\frac{x}{2}\right)}\right)$$

$$\Rightarrow \cot^{-1}\left(\cot\left(\frac{x}{2}\right)\right)$$

But we know that $\cot^{-1}(\cot x) = x$, therefore,

$$\cot^{-1}\left(\sqrt{\frac{1 + \cos x}{1 - \cos x}}\right) = \cot^{-1}\left(\cot\left(\frac{x}{2}\right)\right)$$

$$\Rightarrow \frac{x}{2}$$

Let it be known as equation (1)

Now let us solve $\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$ (2)

Using the identities $1 = \sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)$ and $\sin x = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$ in the denominator of (2) we get

$$\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) = \tan^{-1}\left(\frac{\cos x}{\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right) - 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) = \tan^{-1}\left(\frac{\cos x}{\left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right)^2}\right)$$

Let it be known as equation (3).

Now using the identity $\cos x = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)$ in the numerator of (3) we get,

$$\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) = \tan^{-1}\left(\frac{\left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right) \cdot \left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)\right)}{\left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right)^2}\right) \dots\dots(4)$$

Cancelling the $\left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right)$ term from numerator and denominator of (4) we get

$$\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) = \tan^{-1}\left(\frac{\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)}{\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)}\right) \dots\dots(5)$$

Simplifying it further by taking $\cos\left(\frac{x}{2}\right)$ common from both numerator and denominator of (5) we get,

$$\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) = \tan^{-1}\left(\frac{1 + \tan\left(\frac{x}{2}\right)}{1 - \tan\left(\frac{x}{2}\right)}\right) \dots\dots(6)$$

Now using the identity, $\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(x) + \tan^{-1}(y)$ on (6) we get,

$$\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) = \tan^{-1}(1) + \tan^{-1}\left(\tan\left(\frac{x}{2}\right)\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) = \frac{\pi}{4} + \frac{x}{2}$$

Let it be known as equation (7).

Using equations (1) and (7) we get

$$\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) - \cot^{-1}\left(\sqrt{\frac{1 + \cos x}{1 - \cos x}}\right) = \frac{\pi}{4} + \frac{x}{2} - \frac{x}{2}$$

$$\therefore \tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right) - \cot^{-1}\left(\sqrt{\frac{1 + \cos x}{1 - \cos x}}\right) = \frac{\pi}{4}$$

5. Show that

$$\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \sin^{-1}\left(\frac{x}{a}\right) = \cos^{-1}\left(\frac{\sqrt{a^2 - x^2}}{a}\right)$$

Ans: Using the trigonometric substitution $x = a \sin \theta$ in $\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right)$ we

get,

$$\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \tan^{-1}\left(\frac{a \sin \theta}{\sqrt{a^2 - a^2 \sin^2 \theta}}\right) \dots\dots(1)$$

Taking out a common from the denominator of (1) and cancelling out with numerator we get

$$\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \tan^{-1}\left(\frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}\right) \dots\dots(2)$$

Using the trigonometric identity $\sqrt{1 - \sin^2 x} = \cos x$ in the denominator of (2) we get,

$$\begin{aligned} \tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) &= \tan^{-1}\left(\frac{\sin \theta}{\cos \theta}\right) \\ \Rightarrow \tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) &= \tan^{-1}(\tan(\theta)) \end{aligned}$$

Let it be known as equation (3).

Using the result $\tan^{-1}(\tan x) = x$ in (3) we get

$$\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \theta \dots\dots(4)$$

Let us now resubstitute

$$x = a \sin \theta$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{x}{a}\right)$$

In equation (4) we get,

$$\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \sin^{-1}\left(\frac{x}{a}\right) \dots\dots(5)$$

Now, from $x = a \sin \theta$ we get

$\sin \theta = \frac{x}{a}$. Hence, using the trigonometric identity

$\sin^2 \theta + \cos^2 \theta = 1$ we get,

$$\left(\frac{x}{a}\right)^2 + \cos^2 \theta = 1$$

$$\Rightarrow \cos^2 \theta = 1 - \frac{x^2}{a^2}$$

$$\Rightarrow \cos \theta = \sqrt{\frac{a^2 - x^2}{a^2}}$$

Let this be known as equation (6).

Taking $\cos^{-1}$ on both sides of (6) we get,

$$\cos^{-1}(\cos \theta) = \cos^{-1}\left(\frac{\sqrt{a^2 - x^2}}{a}\right)$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{\sqrt{a^2 - x^2}}{a}\right)$$

From equation (4) we get,

$$\tan^{-1}\left(\frac{x}{\sqrt{a^2 - x^2}}\right) = \cos^{-1}\left(\frac{\sqrt{a^2 - x^2}}{a}\right).$$

6. Show that

$$\cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] = \tan^{-1}\left(\frac{300}{161}\right)$$

Ans: Let us first change $\cos^{-1} \frac{8}{17}$ into the form of tan inverse.

$$\text{Let } \cos^{-1} \frac{8}{17} = x$$

Taking cosine on both sides we get

$$\cos\left(\cos^{-1} \frac{8}{17}\right) = \cos x$$

$$\Rightarrow \cos x = \frac{8}{17}$$

Let this be known as equation (1).

Now using the trigonometric identity $\sin^2 \theta = 1 - \cos^2 \theta$ we get,

$$\sin^2 x = 1 - \left(\frac{8}{17}\right)^2$$

$$\Rightarrow \sin x = \frac{15}{17}$$

Let this be known as equation (2).

From (1) and (2),

$$\tan x = \frac{\sin x}{\cos x}$$

$$\Rightarrow \tan x = \frac{15}{8}$$

Let this be known as equation (3).

Taking $\tan^{-1}$ on both sides of equation (3) we get,

$$\tan^{-1}(\tan x) = \tan^{-1}\left(\frac{15}{8}\right)$$

$$\Rightarrow x = \tan^{-1}\left(\frac{15}{8}\right)$$

Let this be known as equation (4).

Similarly let us now change $\sin^{-1} \frac{8}{17}$ into the form of tan inverse.

$$\text{Let } \sin^{-1} \frac{8}{17} = y$$

Taking sine on both sides we get

$$\sin\left(\sin^{-1} \frac{8}{17}\right) = \sin y$$

$$\Rightarrow \sin y = \frac{8}{17}$$

Let this be known as equation (5).

Now using the trigonometric identity $\cos^2 \theta = 1 - \sin^2 \theta$ on (5) we get,

$$\cos^2 y = 1 - \left(\frac{8}{17}\right)^2$$

$$\Rightarrow \cos y = \frac{15}{17}$$

Let this be known as equation (6).

From (5) and (6),

$$\tan y = \frac{\sin y}{\cos y}$$

$$\Rightarrow \tan y = \frac{8}{15}$$

Let this be known as equation (7).

Taking $\tan^{-1}$ on both sides of equation (7) we get,

$$\tan^{-1}(\tan y) = \tan^{-1}\left(\frac{8}{15}\right)$$

$$\Rightarrow y = \tan^{-1}\left(\frac{8}{15}\right)$$

Let this be known as equation (8).

Substituting these values from (4) and (8) in the original equation we get,

$$\begin{aligned} & \cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] \\ &= \cot^{-1}\left[2 \tan\left(\tan^{-1} \frac{15}{8}\right)\right] + \tan^{-1}\left[2 \tan\left(\tan^{-1} \frac{8}{15}\right)\right] \\ \Rightarrow & \cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] = \cot^{-1}\left(\frac{15}{4}\right) + \tan^{-1}\left(\frac{16}{15}\right) \end{aligned}$$

Let this be known as equation (9).

Now using the trigonometric identity $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ on equation (9) we get,

$$\cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] = \cot^{-1}\left(\frac{15}{4}\right) + \frac{\pi}{2} - \cot^{-1}\left(\frac{16}{15}\right)$$

Let this be known as equation (10).

Again, using the trigonometric identity $\cot^{-1}\left(\frac{xy+1}{y-x}\right) = \cot^{-1}(x) - \cot^{-1}(y)$ on (10) we get,

$$\begin{aligned} & \cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] = \frac{\pi}{2} + \cot^{-1}\left(\frac{5}{\frac{16}{15} - \frac{15}{4}}\right) \\ \Rightarrow & \cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] = \frac{\pi}{2} + \cot^{-1}\left(\frac{300}{-161}\right) \end{aligned}$$

Let this be known as equation (11).

Using $\cot^{-1}(x) = -\cot^{-1}(x)$ on (11) we get,

$$\cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] = \frac{\pi}{2} - \cot^{-1}\left(\frac{300}{161}\right) \dots\dots(12)$$

Again, using the trigonometric identity $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ on (12) we get,

$$\cot^{-1}\left[2 \tan\left(\cos^{-1} \frac{8}{17}\right)\right] + \tan^{-1}\left[2 \tan\left(\sin^{-1} \frac{8}{17}\right)\right] = \tan^{-1}\left(\frac{300}{161}\right)$$

Hence proved.

7. Show that $\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \frac{\pi}{4} + \frac{1}{2}\cos^{-1} x^2$

Ans: To solve this question, substitute $x^2 = \cos \theta$, therefore,

$$\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \tan^{-1}\left(\frac{\sqrt{1+\cos \theta} + \sqrt{1-\cos \theta}}{\sqrt{1+\cos \theta} - \sqrt{1-\cos \theta}}\right) \dots\dots(1)$$

Using the trigonometric identities $\sqrt{1+\cos x} = 2\cos\left(\frac{x}{2}\right)$ and

$\sqrt{1-\cos x} = 2\sin\left(\frac{x}{2}\right)$ on equation (1) we get,

$$\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \tan^{-1}\left(\frac{2\cos\left(\frac{\theta}{2}\right) + 2\sin\left(\frac{\theta}{2}\right)}{2\cos\left(\frac{\theta}{2}\right) - 2\sin\left(\frac{\theta}{2}\right)}\right) \dots\dots(2)$$

Taking $2\cos\left(\frac{\theta}{2}\right)$ common from numerator and denominator of (2) we get,

$$\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \tan^{-1}\left(\frac{1 + \tan\left(\frac{\theta}{2}\right)}{1 - \tan\left(\frac{\theta}{2}\right)}\right) \dots\dots(3)$$

Now using the identity, $\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(x) + \tan^{-1}(y)$ on (3) we get,

$$\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \tan^{-1}(1) + \tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right)$$

$$\Rightarrow \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = \frac{\pi}{4} + \frac{\theta}{2}$$

Let this be known as equation (4).

Resubstituting $x^2 = \cos \theta$ in (4) we get,

$$\tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = \frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2$$

Hence Proved.

8. Solve the following for x: $\cot^{-1} 2x + \cot^{-1} 3x = \frac{\pi}{4}$

Ans: To solve this question, use the identity

$$\cot^{-1} \left(\frac{xy-1}{x+y} \right) = \cot^{-1}(x) + \cot^{-1}(y)$$

$$\cot^{-1} 2x + \cot^{-1} 3x = \cot^{-1} \left(\frac{6x^2-1}{5x} \right)$$

Hence,

$$\cot^{-1} \left(\frac{6x^2-1}{5x} \right) = \frac{\pi}{4} \quad \dots(1)$$

Taking cot on both sides we get,

$$\cot \left(\cot^{-1} \left(\frac{6x^2-1}{5x} \right) \right) = \cot \left(\frac{\pi}{4} \right)$$

$$\Rightarrow \frac{6x^2-1}{5x} = \cot \left(\frac{\pi}{4} \right)$$

Let it be known as equation (2).

Solving the equation (2) by substituting the principal value of $\cot \left(\frac{\pi}{4} \right)$ we get

$$6x^2 - 1 = 5x$$

$$\Rightarrow 6x^2 - 5x - 1 = 0$$

Simplifying it further we get,

$$6x^2 + (-6x + x) - 1 = 0$$

$$(6x + 1)(x - 1) = 0$$

$$\Rightarrow x = 1, -\frac{1}{6}$$

But $x = -\frac{1}{6}$ is not possible. Therefore,

$$x = 1.$$

9. Show that $\tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \frac{\pi}{4}$, $m, n > 0$

Ans: To solve this problem use the trigonometric identity,

$$\tan^{-1}\left(\frac{x-y}{1+xy}\right) = \tan^{-1}(x) - \tan^{-1}(y)$$

Hence,

$$\tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \tan^{-1}\left(\frac{\frac{m}{n} - \frac{m-n}{m+n}}{1 + \left(\frac{m}{n}\right)\left(\frac{m-n}{m+n}\right)}\right) \dots\dots(1)$$

Simplifying the fractions of equation (1) we get

$$\tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \tan^{-1}\left(\frac{\frac{m^2 + n^2}{n(m+n)}}{1 + \left(\frac{m^2 - mn}{n(m+n)}\right)}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \tan^{-1}\left(\frac{m^2 + n^2}{m^2 + n^2}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \tan^{-1}(1)$$

Let this be known as equation (2).

But we know that,

$$\tan\left(\frac{\pi}{4}\right) = 1$$

$$\Rightarrow \tan^{-1}(1) = \frac{\pi}{4}$$

Hence from equation (2) it is proved that

$$\tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \frac{\pi}{4}, \quad m, n > 0$$

10. Prove that $\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right] = \frac{x+y}{1-xy}$

Ans: To solve this problem use the substitution,

$$x = \tan \theta$$

$$y = \tan \alpha$$

In the LHS of the given expression.

$$\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right] = \tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-\tan^2\alpha}{1+\tan^2\alpha}\right)\right]$$

Let this be known as equation (1).

Using the trigonometric identity

$$\frac{2\tan\theta}{1+\tan^2\theta} = \sin 2\theta$$

$$\frac{1-\tan^2\alpha}{1+\tan^2\alpha} = \cos 2\alpha$$

From (1) we get,

$$\begin{aligned} \tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right] &= \tan\left[\frac{1}{2}\sin^{-1}(\sin 2\theta) + \frac{1}{2}\cos^{-1}(\cos 2\alpha)\right] \\ \Rightarrow \tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right] &= \tan[\theta + \alpha] \end{aligned}$$

Let this be known as equation (2).

Now using the trigonometric identity, $\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$ on equation (2)

we get,

$$\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right] = \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} \dots\dots(3)$$

Resubstituting $\theta = \tan^{-1}(x)$ and $\alpha = \tan^{-1}(y)$ in equation (3) we get,

$$\tan \left[\frac{1}{2} \sin^{-1} \left(\frac{2x}{1+x^2} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) \right] = \frac{\tan(\tan^{-1} x) + \tan(\tan^{-1} y)}{1 - \tan(\tan^{-1} x) \cdot \tan(\tan^{-1} y)}$$

$$\Rightarrow \tan \left[\frac{1}{2} \sin^{-1} \left(\frac{2x}{1+x^2} \right) + \frac{1}{2} \cos^{-1} \left(\frac{1-y^2}{1+y^2} \right) \right] = \frac{x+y}{1-xy}$$

11. Solve the following for x: $\cos^{-1} \left(\frac{x^2-1}{x^2+1} \right) + \frac{1}{2} \tan^{-1} \left(\frac{-2x}{1-x^2} \right) = \frac{2\pi}{3}$

Ans: To solve this question, the substitution $x = \tan \theta$ in the LHS of the given equation

$$\cos^{-1} \left(\frac{x^2-1}{x^2+1} \right) + \frac{1}{2} \tan^{-1} \left(\frac{-2x}{1-x^2} \right) = \cos^{-1} \left(\frac{\tan^2 \theta - 1}{\tan^2 \theta + 1} \right) + \frac{1}{2} \tan^{-1} \left(\frac{-2 \tan \theta}{1 - \tan^2 \theta} \right)$$

Let this be known as equation (1).

Using the trigonometric identities $\frac{2 \tan \theta}{1 - \tan^2 \theta} = \tan 2\theta$ and $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos 2\theta$

From (1) we get,

$$\cos^{-1} \left(\frac{x^2-1}{x^2+1} \right) + \frac{1}{2} \tan^{-1} \left(\frac{-2x}{1-x^2} \right) = \cos^{-1}(-\cos 2\theta) + \frac{1}{2} \tan^{-1}(-\tan 2\theta) \dots\dots(2)$$

But,

$$\cos^{-1}(-x) = \pi - \cos^{-1}(x)$$

$$\tan^{-1}(-x) = -\tan^{-1}(x)$$

Therefore from (2) we get,

$$\cos^{-1} \left(\frac{x^2-1}{x^2+1} \right) + \frac{1}{2} \tan^{-1} \left(\frac{-2x}{1-x^2} \right) = \pi - \cos^{-1}(\cos 2\theta) - \frac{1}{2} \tan^{-1}(\tan 2\theta)$$

$$\Rightarrow \cos^{-1} \left(\frac{x^2-1}{x^2+1} \right) + \frac{1}{2} \tan^{-1} \left(\frac{-2x}{1-x^2} \right) = \pi - 2\theta - \theta$$

$$\Rightarrow \cos^{-1} \left(\frac{x^2-1}{x^2+1} \right) + \frac{1}{2} \tan^{-1} \left(\frac{-2x}{1-x^2} \right) = \pi - 3\theta$$

Let this be known as equation (3).

Re substituting $\theta = \tan^{-1}(x)$ in equation (3) we get,

$$\cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) + \frac{1}{2}\tan^{-1}\left(\frac{-2x}{1-x^2}\right) = \pi - 3\tan^{-1}x \quad \dots\dots(4)$$

But it is given that $\cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) + \frac{1}{2}\tan^{-1}\left(\frac{-2x}{1-x^2}\right) = \frac{2\pi}{3}$. Therefore from (4)

we get,

$$\pi - 3\tan^{-1}x = \frac{2\pi}{3}$$

$$\Rightarrow 3\tan^{-1}x = \frac{\pi}{3}$$

$$\Rightarrow x = \tan\left(\frac{\pi}{9}\right)$$

12. Prove that $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{8}\right) = \frac{\pi}{4}$

Ans: To solve this problem use the trigonometric identity,

$$\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(x) + \tan^{-1}(y) \text{ in the LHS of the given expression}$$

Hence,

$$\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{8}\right) = \tan^{-1}\left(\frac{\frac{1}{3} + \frac{1}{5}}{1 - \frac{1}{15}}\right) + \tan^{-1}\left(\frac{\frac{1}{7} + \frac{1}{8}}{1 - \frac{1}{56}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{8}\right) = \tan^{-1}\left(\frac{4}{7}\right) + \tan^{-1}\left(\frac{3}{11}\right)$$

Let this be known as equation (1).

Again, using the trigonometric identity $\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(x) + \tan^{-1}(y)$ on

(1) we get,

$$\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{8}\right) = \tan^{-1}\left(\frac{\frac{4}{7} + \frac{3}{11}}{1 - \frac{12}{77}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{8}\right) = \tan^{-1}\left(\frac{65}{65}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{8}\right) = \tan^{-1}(1)$$

Let this be known as equation (2).

But we know that,

$$\tan\left(\frac{\pi}{4}\right) = 1$$

$$\Rightarrow \tan^{-1}(1) = \frac{\pi}{4}$$

Hence from equation (2) it is proved that

$$\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{8}\right) = \frac{\pi}{4}$$

13. Solve the following for x: $\tan(\cos^{-1} x) = \sin(\tan^{-1} 2)$, $x > 0$

Ans: To solve this question, write $\cos^{-1} x$ in LHS of the given expression in terms of tan inverse. Substitute,

$$\cos^{-1} x = \theta$$

$$\Rightarrow \cos \theta = x$$

Let this be known as equation (1).

Now using the trigonometric identity $\sin \theta = \sqrt{1 - \cos^2 \theta}$ we get from (1),

$$\sin \theta = \sqrt{1 - x^2} \dots\dots(2)$$

Hence from (1) and (2) we get,

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\Rightarrow \tan \theta = \frac{\sqrt{1 - x^2}}{x}$$

Let this be known as equation (3).

Taking tan inverse on both sides of equation (3) we get

$$\tan^{-1}(\tan \theta) = \tan^{-1}\left(\frac{\sqrt{1 - x^2}}{x}\right)$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{\sqrt{1 - x^2}}{x}\right)$$

Let this be known as equation (4).

Hence from (4), we get the given equation as

$$\tan \left(\tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right) \right) = \sin \left(\tan^{-1} 2 \right)$$

$$\Rightarrow \frac{\sqrt{1-x^2}}{x} = \sin \left(\tan^{-1} 2 \right)$$

Let this be known as equation (5).

Now again write $\tan^{-1} 2$ from RHS of equation (5) in terms of sine inverse.

Substitute,

$$\tan^{-1} 2 = y$$

$$\Rightarrow \tan y = 2$$

Let this be known as equation (6).

Now using the trigonometric identity $\sqrt{1+\tan^2 \theta} = \sec \theta$ from (6) we get,

$$\sec y = \sqrt{1+(2)^2}$$

$$\Rightarrow \sec y = \sqrt{5}$$

$$\Rightarrow \cos y = \frac{1}{\sqrt{5}}$$

Let this be known as equation (7).

Using the trigonometric identity $\sin \theta = \sqrt{1-\cos^2 \theta}$ we get from (7),

$$\sin y = \sqrt{1-\left(\frac{1}{\sqrt{5}}\right)^2}$$

$$\Rightarrow \sin y = \frac{2}{\sqrt{5}}$$

$$\Rightarrow y = \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$$

Let this be known as equation (8).

Hence from equations (5), (6) and (8) we get,

$$\frac{\sqrt{1-x^2}}{x} = \frac{2}{\sqrt{5}} \quad \dots\dots(9)$$

Squaring both sides and cross multiplying we get,

$$5(1-x^2) = 4x^2$$

$$\Rightarrow 5 = 9x^2$$

$$\Rightarrow x = \frac{\sqrt{5}}{3}$$

(Since $x > 0$)

14. Prove that $2 \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{4}\right) = \tan^{-1}\left(\frac{32}{43}\right)$

Ans: To solve this question, use the trigonometric identity,

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) = 2 \tan^{-1}(x) \text{ in the LHS of the given expression.}$$

$$2 \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{4}\right) = \tan^{-1}\left(\frac{2\left(\frac{1}{5}\right)}{1-\left(\frac{1}{5}\right)^2}\right) + \tan^{-1}\left(\frac{1}{4}\right)$$

$$\Rightarrow 2 \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{4}\right) = \tan^{-1}\left(\frac{5}{12}\right) + \tan^{-1}\left(\frac{1}{4}\right)$$

Let this be known as equation (1).

Now use the trigonometric identity, $\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(x) + \tan^{-1}(y)$ on equation (1) we get,

$$2 \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{4}\right) = \tan^{-1}\left(\frac{\frac{5}{12} + \frac{1}{4}}{1 - \frac{5}{48}}\right)$$

$$\Rightarrow 2 \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{4}\right) = \tan^{-1}\left(\frac{32}{43}\right)$$

15. Evaluate $\tan\left[\frac{1}{2} \cos^{-1}\left(\frac{3}{\sqrt{11}}\right)\right]$

Ans: To solve this question, use the substitution $\frac{1}{2} \cos^{-1}\left(\frac{3}{\sqrt{11}}\right) = x$ in the LHS of the given expression. Therefore,

$$\frac{1}{2} \cos^{-1}\left(\frac{3}{\sqrt{11}}\right) = x$$

$$\Rightarrow \cos 2x = \frac{3}{\sqrt{11}}$$

Let this be known as equation (1).

But we know that $\frac{1 - \tan^2 x}{1 + \tan^2 x} = \cos 2x$. Hence from (1) we get,

$$\frac{1 - \tan^2 x}{1 + \tan^2 x} = \frac{3}{\sqrt{11}} \quad \dots\dots(2)$$

Applying the rule of component and divided on equation (2) we get,

$$\begin{aligned} \frac{1 - \tan^2 x + 1 + \tan^2 x}{1 - \tan^2 x - 1 - \tan^2 x} &= \frac{3 + \sqrt{11}}{3 - \sqrt{11}} \\ \Rightarrow \frac{2}{-2 \tan^2 x} &= \frac{3 + \sqrt{11}}{3 - \sqrt{11}} \end{aligned}$$

Let this be known as equation (3).

Taking reciprocal of equation (3) we get,

$$\begin{aligned} -\tan^2 x &= \frac{3 - \sqrt{11}}{3 + \sqrt{11}} \\ \Rightarrow \tan^2 x &= \frac{\sqrt{11} - 3}{3 + \sqrt{11}} \\ \Rightarrow \tan x &= \sqrt{\frac{\sqrt{11} - 3}{3 + \sqrt{11}}} \end{aligned}$$

Let this be known as equation (4).

Now resubstituting $\frac{1}{2} \cos^{-1} \left(\frac{3}{\sqrt{11}} \right) = x$ in (4) we get,

$$\tan \left[\frac{1}{2} \cos^{-1} \left(\frac{3}{\sqrt{11}} \right) \right] = \sqrt{\frac{\sqrt{11} - 3}{3 + \sqrt{11}}}$$

16. Prove that $\tan^{-1} \left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right) = \tan^{-1} \left(\frac{a}{b} \right) - x$

Ans: To solve this question, take $b \cos x$ common from both numerator and denominator from the LHS of the given equation.

$$\tan^{-1} \left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right) = \tan^{-1} \left(\frac{\frac{a}{b} - \tan x}{1 + \frac{a}{b} \tan x} \right) \quad \dots\dots(1)$$

Now use the trigonometric identity, $\tan^{-1}\left(\frac{x-y}{1+xy}\right) = \tan^{-1}(x) - \tan^{-1}(y)$ on

equation (1) we get,

$$\begin{aligned}\tan^{-1}\left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x}\right) &= \tan^{-1}\left(\frac{a}{b}\right) - \tan^{-1}(\tan x) \\ \Rightarrow \tan^{-1}\left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x}\right) &= \tan^{-1}\left(\frac{a}{b}\right) - x\end{aligned}$$

17. Prove that

$$\cot\left\{\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right\} + \cos^{-1}(1-2x^2) + \cos^{-1}(2x^2-1) = \pi, \quad x > 0$$

Ans: To solve this question, use the trigonometric identity $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$

on $\cot\left\{\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right\}$. Therefore,

$$\begin{aligned}\cot\left\{\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right\} &= \cot\left\{\frac{\pi}{2} - \cot^{-1}x + \frac{\pi}{2} - \cot^{-1}\left(\frac{1}{x}\right)\right\} \\ \Rightarrow \cot\left\{\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right\} &= \cot\left\{\pi - \left[\cot^{-1}x + \cot^{-1}\left(\frac{1}{x}\right)\right]\right\}\end{aligned}$$

Let this be known as equation (1).

Now using the trigonometric identity $\cot^{-1}\left(\frac{xy-1}{x+y}\right) = \cot^{-1}(x) + \cot^{-1}(y)$ on equation (1) we get,

$$\begin{aligned}\cot\left\{\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right\} &= \cot\left\{\pi - \cot^{-1}\left(\frac{x \cdot \frac{1}{x} - 1}{x + \frac{1}{x}}\right)\right\} \\ \Rightarrow \left\{\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right\} &= \cot\left\{\pi - \left[\cot^{-1}(0)\right]\right\}\end{aligned}$$

Let this be known as equation (2).

But we know that the principal value of

$$\cot\left(\frac{\pi}{2}\right) = 0$$

$$\Rightarrow \cot^{-1}(0) = \frac{\pi}{2}$$

Hence, from equation (2) we get,

$$\cot \left\{ \tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right) \right\} = \cot \left(\frac{\pi}{2} \right)$$

$$\Rightarrow \cot \left\{ \tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right) \right\} = 0$$

Let this be known as equation (3).

Also, we know that $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$. Therefore

$$\cos^{-1}(1 - 2x^2) + \cos^{-1}(2x^2 - 1) = \cos^{-1}(1 - 2x^2) + \cos^{-1}(-(-2x^2 + 1))$$

$$\Rightarrow \cos^{-1}(1 - 2x^2) + \cos^{-1}(2x^2 - 1) = \cos^{-1}(1 - 2x^2) + \pi - \cos^{-1}(2x^2 - 1)$$

$$\Rightarrow \cos^{-1}(1 - 2x^2) + \cos^{-1}(2x^2 - 1) = \pi$$

Let this be known as equation (4).

Hence from equations (3) and (4) we get

$$\cot \left\{ \tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right) \right\} + \cos^{-1}(1 - 2x^2) + \cos^{-1}(2x^2 - 1) = \pi$$

18. Prove that $\tan^{-1} \left(\frac{a-b}{1+ab} \right) + \tan^{-1} \left(\frac{b-c}{1+bc} \right) + \tan^{-1} \left(\frac{c-a}{1+ac} \right) = 0$ where

$a, b, c > 0$

Ans: To solve this question, use the trigonometric identity,

$$\tan^{-1} \left(\frac{x-y}{1+xy} \right) = \tan^{-1}(x) - \tan^{-1}(y) \text{ on the LHS of the given expression.}$$

Therefore,

$$\begin{aligned} & \tan^{-1} \left(\frac{a-b}{1+ab} \right) + \tan^{-1} \left(\frac{b-c}{1+bc} \right) + \tan^{-1} \left(\frac{c-a}{1+ac} \right) \\ &= \left(\tan^{-1}(a) - \tan^{-1}(b) \right) + \left(\tan^{-1}(b) - \tan^{-1}(c) \right) + \left(\tan^{-1}(c) - \tan^{-1}(a) \right) \\ &= 0 \end{aligned}$$

19. Solve the following for x : $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$

Ans: To solve this question, use the trigonometric identity,

$$2 \tan^{-1}(x) = \tan^{-1} \left(\frac{2x}{1-x^2} \right) \text{ on the LHS of the given expression}$$

$$2 \tan^{-1}(\cos x) = \tan^{-1} \left(\frac{2 \cos x}{1 - \cos^2 x} \right) \dots\dots(1)$$

Using the trigonometric identity $\sin^2 x = 1 - \cos^2 x$ in the denominator of (1) we get,

$$2 \tan^{-1}(\cos x) = \tan^{-1}\left(\frac{2 \cos x}{\sin^2 x}\right) \dots\dots(2)$$

Substituting the value from equation (2) into the given expression we get,

$$\tan^{-1}\left(\frac{2 \cos x}{\sin^2 x}\right) = \tan^{-1}(2 \operatorname{cosec} x) \dots\dots(3)$$

Taking tan on both the sides of equation (3) we get,

$$\tan\left(\tan^{-1}\left(\frac{2 \cos x}{\sin^2 x}\right)\right) = \tan\left(\tan^{-1}(2 \operatorname{cosec} x)\right)$$

$$\Rightarrow \frac{2 \cos x}{\sin^2 x} = 2 \operatorname{cosec} x$$

$$\Rightarrow \frac{\cos x}{\sin^2 x} = \frac{1}{\sin x}$$

$$\Rightarrow \cos x = \sin x$$

it is possible only when $x = \frac{\pi}{4}$.

20. Express $\sin^{-1}\left(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right)$ in the simplest form.

Ans: To solve this question, use the substitution $x = \sin \theta$ and $\sqrt{x} = \sin \phi$.
Therefore,

$$\sin^{-1}\left(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right) = \sin^{-1}\left(\sin \theta \sqrt{1-\sin^2 \phi} - \sin \phi \sqrt{1-\sin^2 \theta}\right)$$

$$\Rightarrow \sin^{-1}\left(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right) = \sin^{-1}(\sin \theta \cos \phi - \sin \phi \cos \theta)$$

Let this be known as equation (1).

But we know that $\sin a - \sin b = \sin a \cos b - \cos a \sin b$. Hence from (1) we get,

$$\sin^{-1}\left(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right) = \sin^{-1}(\sin(\theta - \phi))$$

$$\Rightarrow \sin^{-1}\left(x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right) = \theta - \phi$$

Let this be known as equation (2).

Resubstituting $x = \sin \theta$ and $\sqrt{x} = \sin \phi$ in equation (2) we get,

$$\sin^{-1}\left(x\sqrt{1-x}-\sqrt{x}\sqrt{1-x^2}\right)=\sin^{-1}(x)-\sin^{-1}(\sqrt{x})$$

21. If $\tan^{-1}(a) + \tan^{-1}(b) + \tan^{-1}(c) = \pi$ then prove that $a + b + c = abc$.

Ans: To solve this question, use the trigonometric identity,

$$\tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}(x) + \tan^{-1}(y) \text{ on the LHS of the given expression.}$$

Therefore,

$$\tan^{-1}(a) + \tan^{-1}(b) = \tan^{-1}\left(\frac{a+b}{1-ab}\right) \dots\dots(1)$$

Again, using the same trigonometric identity on the LHS of the given expression and using (1), we get,

$$\tan^{-1}(a) + \tan^{-1}(b) + \tan^{-1}(c) = \tan^{-1}\left(\frac{a+b}{1-ab}\right) + \tan^{-1}(c)$$

$$\Rightarrow \tan^{-1}(a) + \tan^{-1}(b) + \tan^{-1}(c) = \tan^{-1}\left(\frac{\frac{a+b}{1-ab} + c}{1 - \frac{c(a+b)}{1-ab}}\right)$$

$$\Rightarrow \tan^{-1}(a) + \tan^{-1}(b) + \tan^{-1}(c) = \tan^{-1}\left(\frac{a+b+c-abc}{1-ab-ca-bc}\right)$$

Let this be known as equation (2).

But it is given that $\tan^{-1}(a) + \tan^{-1}(b) + \tan^{-1}(c) = \pi$. Therefore, from equation (2) we get,

$$\tan^{-1}\left(\frac{a+b+c-abc}{1-ab-ca-bc}\right) = \pi \dots\dots(3)$$

Taking tan on both sides of the equation (3) we get,

$$\tan\left(\tan^{-1}\left(\frac{a+b+c-abc}{1-ab-ca-bc}\right)\right) = \tan(\pi)$$

$$\Rightarrow \frac{a+b+c-abc}{1-ab-ca-bc} = 0$$

$$\Rightarrow a+b+c-abc = 0$$

Hence proved that if $\tan^{-1}(a) + \tan^{-1}(b) + \tan^{-1}(c) = \pi$ then $a + b + c = abc$.

22. If $\sin^{-1}x > \cos^{-1}x$ then x belong to which category?

Ans: To solve this question, use the trigonometric identity,

$$\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}.$$

$$\sin^{-1} x > \cos^{-1} x$$

$$\Rightarrow \sin^{-1} x > \frac{\pi}{2} - \sin^{-1} x$$

$$\Rightarrow 2\sin^{-1} x > \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} x > \frac{\pi}{4}$$

Let this be known as expression (1).

Taking sine on both the sides of expression (1) we get,

$$\sin(\sin^{-1} x) > \sin\left(\frac{\pi}{4}\right)$$

$$\Rightarrow x > \sin\left(\frac{\pi}{4}\right)$$

$$\Rightarrow x > \frac{1}{\sqrt{2}}$$

Let this be known as expression (2).

Also, we know that $-1 \leq x \leq 1$ (since range of sin and cos is $[-1, 1]$)(3)

Hence from (2) and (3), the interval in which x belongs is $\left(\frac{1}{\sqrt{2}}, 1\right)$.